Main Thesis of GenBrain

Cardiology made major strides when an analogy was drawn between the heart and a mechanical pump. Descartes, Harvey, Hales, and others began to understand the circulatory system by watching how a bellows moves air or how piston-based hydraulic systems like fountains or firefighting systems moved water from place to place. This critical analogy eventually allowed for diagnoses using pump terminology ("valves", "ejection fraction", "blood pressure"), and predictions about heart outcomes given analagous insults to a mechanical device (i.e. making a hole in a pump tube will induce the same outcome as a vascular perforation; clogging the tubes of the pump requires more force, as in hypertension). Nearly every aspect of cardiology is now framed in this view, and as any good model should, the pump model opened doors for framing questions and experiments, understanding the heart's failure modes, and designing effective interventions and treatments.

Unfortunately, there is no useful analogy that has allowed us to develop a similar multi-level understanding of the brain. We currently describe mental health phenomena at the level of behavioral and thought patterns; these patterns can be observed and categorized in order to prescribe treatments that generally *stabilize* affected individuals, but are largely ineffective at halting recurring thought disorders. The organ that generates these thinking patterns remains a mystery: comparison between the brains of individuals with schizophrenia and those without provides little insight, because brain "abnormalities" that are associated with the disorder (e.g. ventricle size, frontal cortex size, hippocampus size) fall within the range of normal human variance. Genetic abnormalities have likewise come up empty, again with most genetic anomalies observed consistently in the healthy human population. Without a reference point to gauge how brains actually work, and without a clear smoking gun from non-computational hypotheses like anatomy and genetics, we should turn to cardiology's example and answer "What is the pump of the brain?" This is a difficult question because the analogy is likely 3-pronged (see below), and our current best analogies fail to provide inroads to understanding. For example, the "digital computer" analogy fails because, unlike computers:

1. The hardware of the brain is not fixed. It is constantly rewiring itself and changing throughout development to adapt to new environments and behavioral demands
2. The brain operates on about 10 watts of power and individual units operate at a max of ~500Hz: GPU-based computers consume about two orders of magnitude more power, with units operating in the GHz range.
3. The brain is aware of itself and other agents. It undergoes subjective experience, moods, and emotions.
4. The brain learns mostly in an undirected environment without clear rules, and generalizes through analogy.

Recently, deep neural networks have become a popular analogy for brains due to their adaptibility, their use of neural-like elements, and their extraordinary performance on human-inspired tasks including language and vision. Some of our collaborators have taken the analogy quite